

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
saidkurbanov@saidkurbanov:~$ ls /etc/ > file.txt
saidkurbanov@saidkurbanov:~$ ls >> file.txt
saidkurbanov@saidkurbanov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
```

Рисунок 1: Запись в файл

```
saidkurbanov@saidkurbanov:~$ grep .conf file.txt > conf.txt
saidkurbanov@saidkurbanov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

Рисунок 2: Поиск расширения .conf


```
-personal/stage03/report/_resources/csl
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage03/report/bib/cite.bib
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/_resources/csl
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/bib/cite.bib
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/_resources/csl
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/bib/cite.bib
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/_resources/csl
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/bib/cite.bib
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_resources/csl
/home/saidkurbanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib/cite.bib
/home/saidkurbanov/git-extended/.git/hooks/commit-msg.sample
/home/saidkurbanov/git-extended/.git/objects/d4/c801fe194bba0c1c878a0c879a97fcf030b61e
/home/saidkurbanov/git-extended/.git/objects/c7
/home/saidkurbanov/git-extended/.git/objects/ca
/home/saidkurbanov/git-extended/.git/config
/home/saidkurbanov/conf.txt
saidkurbanov@saidkurbanov:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new  
find: '/etc/libvirt': Permission denied  
/etc/hp  
/etc/hp/hplip.conf  
/etc/httpd  
/etc/httpd/conf/httpd.conf  
/etc/libibverbs.d/hfi1verbs.driver  
/etc/libibverbs.d/hns.driver  
find: '/etc/lvm/archive': Permission denied  
/etc/logrotate.d/httpd  
find: '/etc/lvm/backup': Permission denied  
find: '/etc/lvm/cache': Permission denied  
find: '/etc/lvm/devices': Permission denied  
find: '/etc/nftables': Permission denied  
find: '/etc/openvpn/client': Permission denied  
find: /etc/nvme/hostnqn  
/etc/nvme/hostid  
'/etc/openvpn/server': Permission denied  
find: '/etc/pki/rsyslog': Permission denied  
find: '/etc/polkit-1/localauthority': Permission denied  
find: '/etc/polkit-1/rules.d': Permission denied  
find: '/etc/sos/cleaner': Permission denied  
/etc/sane.d/dll.d/hpaio  
/etc/sane.d/hp.conf  
/etc/sane.d/hp3900.conf  
/etc/sane.d/hp4200.conf  
/etc/sane.d/hp5400.conf  
/etc/sane.d/hpsj5s.conf  
:find: '/etc/ssh/sshd_config.d': Permission denied  
find: '/etc/sss': Permission denied  
find: '/etc/sudoers.d': Permission denied  
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
saidkurbanov@saidkurbanov:~$  
saidkurbanov@saidkurbanov:~$ find ~ -name "log*" > logfile &  
[1] 6572  
saidkurbanov@saidkurbanov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
saidkurbanov@saidkurbanov:~$  
saidkurbanov@saidkurbanov:~$ rm logfile  
saidkurbanov@saidkurbanov:~$
```

Рисунок 5: Фоновый запуск процесса

```
saidkurbanov@saidkurbanov:~$  
saidkurbanov@saidkurbanov:~$ gedit &  
[1] 6616  
saidkurbanov@saidkurbanov:~$ ps | grep gedit  
    6616 pts/0    00:00:00 gedit  
saidkurbanov@saidkurbanov:~$ kill 6616  
[1]+  Terminated                  gedit  
saidkurbanov@saidkurbanov:~$  
saidkurbanov@saidkurbanov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space
    available on the file system containing each file name argument.  If no file name is
    given, the space available on all currently mounted file systems is shown.  Space is
    shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is
    set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file
    system, df shows the space available on that file system rather than on the file
    system containing the device node.  This version of df cannot show the space avail-
    able on unmounted file systems, because on most kinds of systems doing so requires
    non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file sys-
    tems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                User Commands                                du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is
        usually smaller, it may be larger due to holes in ('sparse') files, internal
        fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
        of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
saidkurbanov@saidkurbanov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89764064  64759360  59% /
devtmpfs         4011292      0    4011292   0% /dev
tmpfs            4035668      92    4035576   1% /dev/shm
tmpfs            1614268     1952    1612316   1% /run
tmpfs             1024        0        1024   0% /run/credentials/systemd-journald.service
tmpfs            4035668     1612    4034056   1% /tmp
/dev/nvme0n1p3 155186176 89764064  64759360  59% /home
/dev/nvme0n1p2  1992552     518800    1352512  28% /boot
/dev/loop0        68480      68480          0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2        49280      49280          0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop1       115712     115712          0 100% /var/lib/snapd/snap/hugo/25676
tmpfs             1024        0        1024   0% /run/credentials/systemd-resolved.service
tmpfs            807132      164    806968   1% /run/user/1033
/dev/sr0         2677920    2677920          0 100% /run/media/saidkurbanov/Fedora-WS-Live-43
saidkurbanov@saidkurbanov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/3c
4      ./git-extended/.git/objects/c7
4      ./git-extended/.git/objects/02
4      ./git-extended/.git/objects/ca
4      ./git-extended/.git/objects/8c
4      ./git-extended/.git/objects/9e
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
829676 .
saidkurbanov@saidkurbanov:~$
```

Рисунок 10: Запуск команды du


```
/home/saidkurbanov/git-extended/.git/objects/d1
/home/saidkurbanov/git-extended/.git/objects/fb
/home/saidkurbanov/git-extended/.git/objects/08
/home/saidkurbanov/git-extended/.git/objects/34
/home/saidkurbanov/git-extended/.git/objects/3c
/home/saidkurbanov/git-extended/.git/objects/c7
/home/saidkurbanov/git-extended/.git/objects/02
/home/saidkurbanov/git-extended/.git/objects/ca
/home/saidkurbanov/git-extended/.git/objects/8c
/home/saidkurbanov/git-extended/.git/objects/9e
/home/saidkurbanov/git-extended/.git/logs
/home/saidkurbanov/git-extended/.git/logs/refs
/home/saidkurbanov/git-extended/.git/logs/refs/heads
/home/saidkurbanov/git-extended/.git/logs/refs/remotes
/home/saidkurbanov/git-extended/.git/logs/refs/remotes/origin
/home/saidkurbanov/monthly
/home/saidkurbanov/reports
/home/saidkurbanov/reports/monthly
/home/saidkurbanov/reports/monthly/monthly
/home/saidkurbanov/ski.plases
/home/saidkurbanov/ski.plases/equipment
/home/saidkurbanov/ski.plases/plans
/home/saidkurbanov/australia
/home/saidkurbanov/play
/home/saidkurbanov/play/games
/home/saidkurbanov/play/games/play
saidkurbanov@saidkurbanov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.